Operations Research 1: Deterministic Models
Nonlinear Programming

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Final Exam: Classroom Allocation

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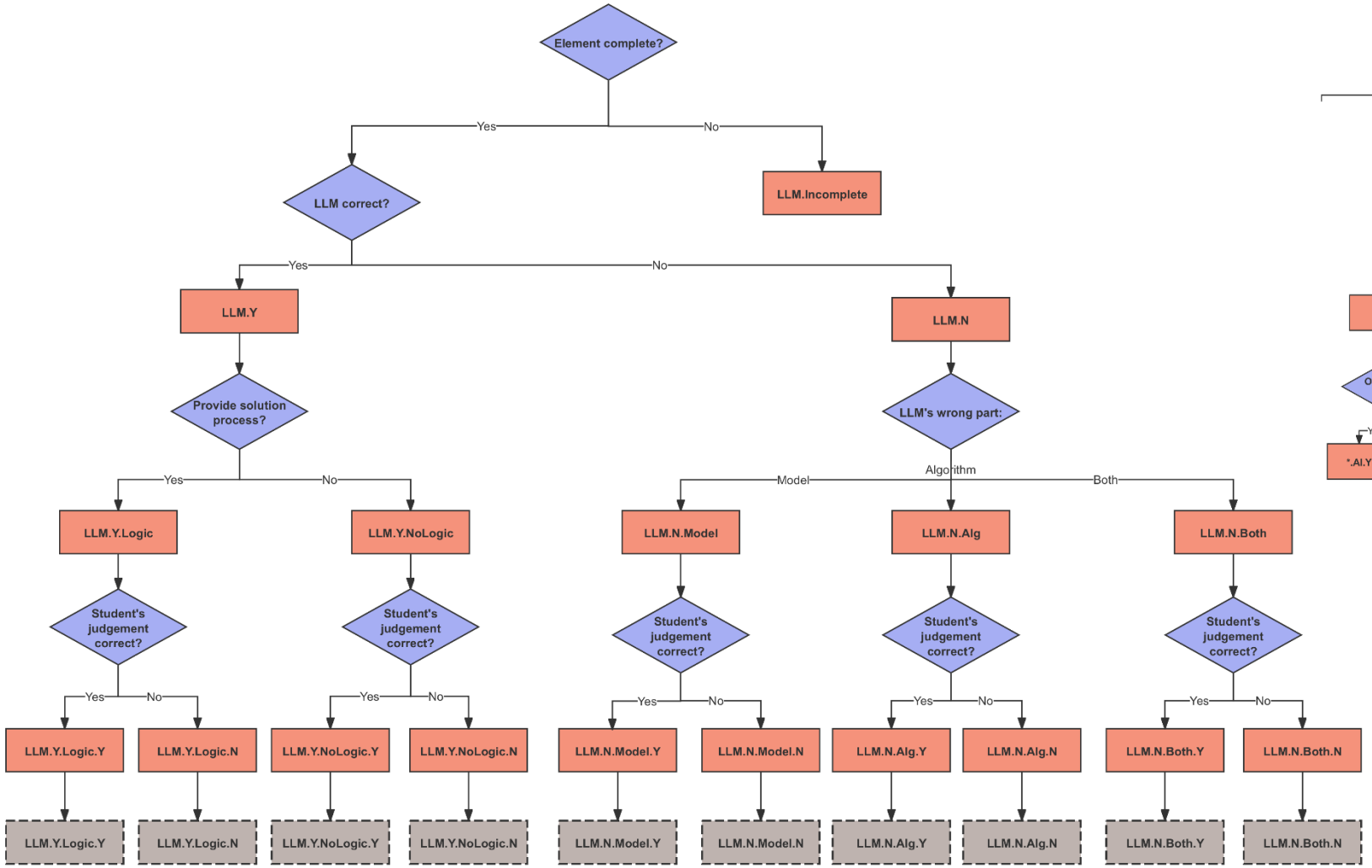
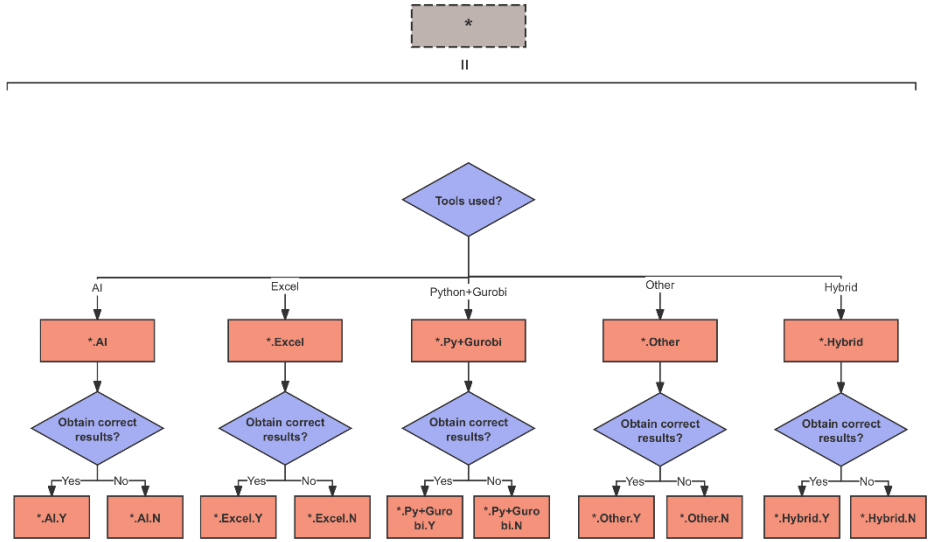
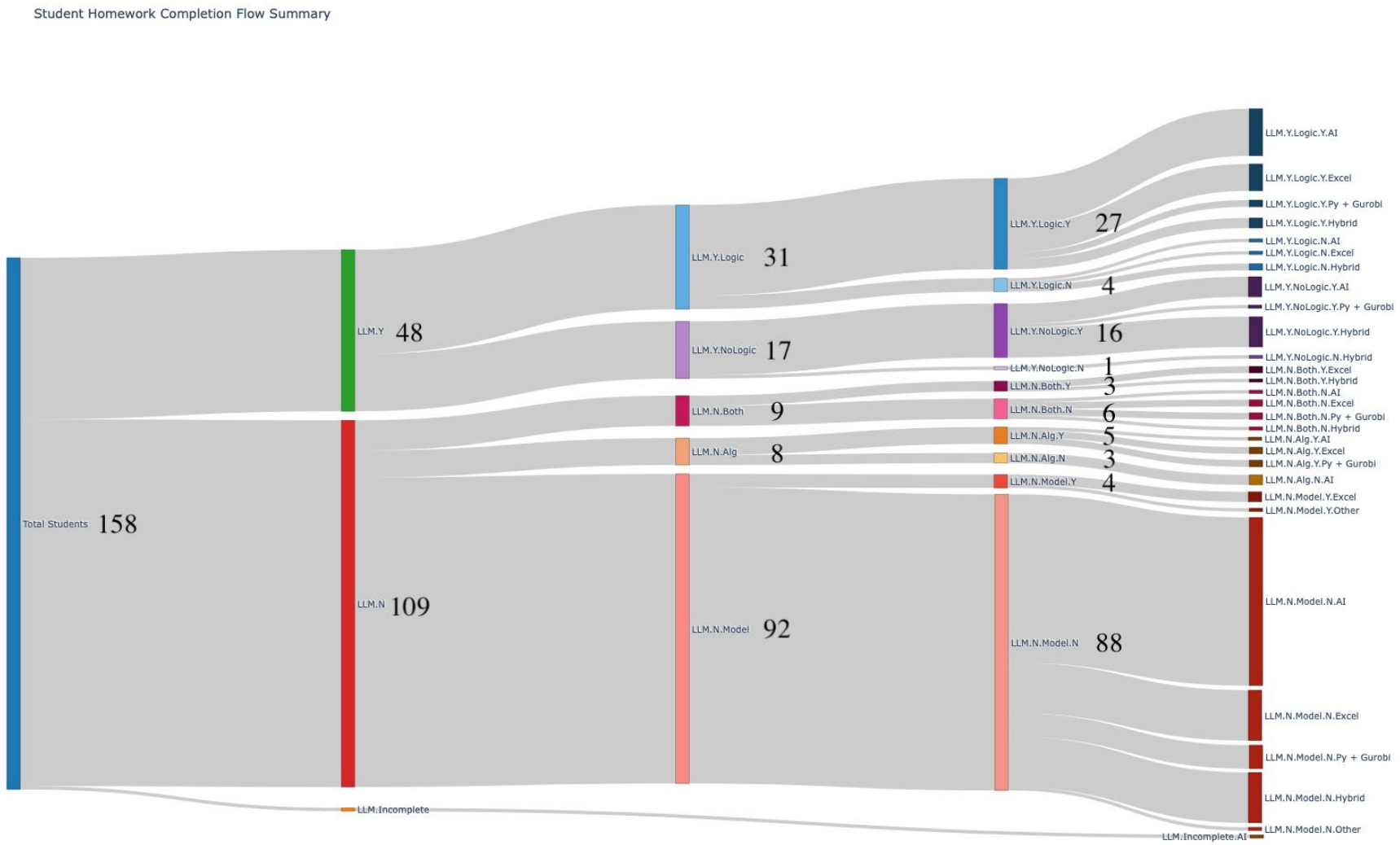
Homework 08: IP Modeling (with AI Tools)

Homework 08: Flow Chart





Homework 08: Sankey Diagram



Wyndor Example (Revisit)

Wyndor Glass Co.

- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
- Three plants
  - Plant 1
    - Aluminum frames and hardware
  - Plant 2
    - Wood frames
  - Plant 3
    - Glass and product assemble
- Data
  - Available production time in each plant (hours / week)
  - Production time used in each plant for each batch of each product (hours / batch)
  - Profit per batch of each product

	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- Decision: production
- Objective: to maximize the total profit (per week)

Wyndor Glass Co. (Revisit)

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \leq 4, \\
& 2x_2 \leq 12, \\
& 3x_1 + 2x_2 \leq 18, \\
& x_1, \quad x_2 \geq 0.
\end{aligned}
$$

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2	<=	4
Plant 2	0	2	12	<=	12
Plant 3	3	2	18	<=	18
	Doors	Windows			Total profit
Batches produced	2	6			$36,000

Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Initial price	$3,000	$5,000			
Unit discount	100	300			
Profit per batch					
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{array}{lrrl}
\max & z = (3-0.1x_1)x_1 + (5-0.3x_2)x_2 & & \\
\text{s.t.} & x_1 & & \leq 4, \\
 & & 2x_2 & \leq 12, \\
 & 3x_1 & +2x_2 & \leq 18, \\
 & x_1, & x_2 & \geq 0.
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	Doors	Windows			
Initial price	$3,000	$5,000			
Unit discount	100	300			
Profit per batch					
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

	Doors	Windows			
Initial price	$3,000	$5,000			
Unit discount	100	300			
Profit per batch	$2,768	$3,345			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2.32	<=	4
Plant 2	0	2	11.03	<=	12
Plant 3	3	2	18.00	<=	18
	Doors	Windows			Total profit
Batches produced	2.32	5.52			$24,881

Four Assumptions of LP (Revisit)

- Proportionality
  - Objective value/LHS is proportional to the level of activity
  - Nonlinear/mixed integer programming
- Additivity
  - Sum of individual contributions (objective/LHS)
  - Nonlinear programming
- Divisibility
  - (Mixed) Integer programming
- Certainty
  - Sensitivity analysis, but …
  - Stochastic programming, (approximate) dynamic programming, simulation optimization, robust optimization, …

Basic Concepts

General Form

- Consider the following optimization problem

$$\begin{array}{lll} \min & f(x) & & \text{(objective function)} \\ \text{s.t.} & g_i(x) \le 0, & i = 1, \ldots, m & \text{(inequality constraints)} \\ & h_i(x) = 0, & i = 1, \ldots, l & \text{(equality constraints)} \\ & x \in X. & & \text{(set constraints)} \end{array}$$

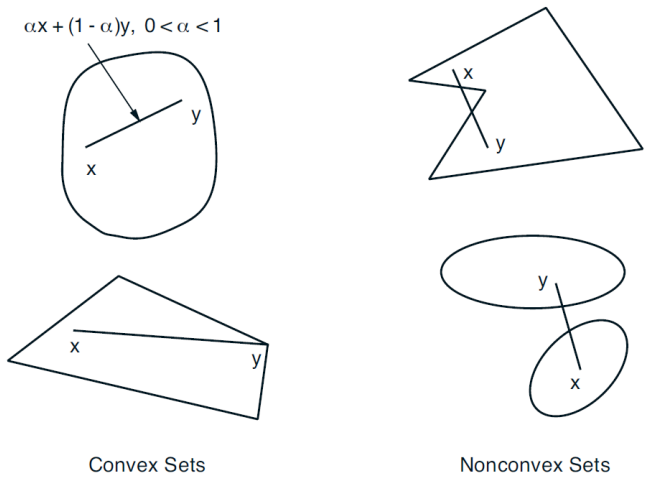
  - X: lower and upper bounds on x, or specially structured constraints

- Unconstrained optimization

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in R^n. \end{array}$$

Convex Sets

- The set $S \subset R^n$ is a convex set if for each $x, y \in S$ and for each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \in S$.



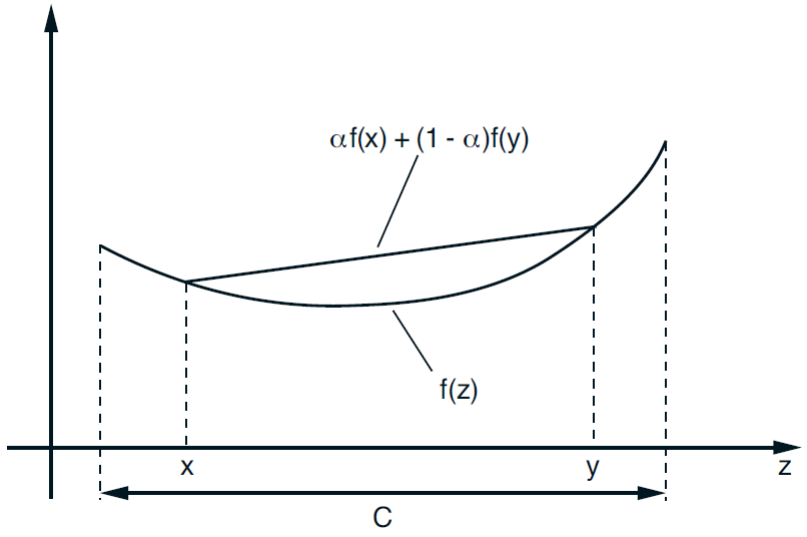
- Examples
  - Hyperplane: $S = \{x \in R^n : a^T x = k\}$
  - Half-space: $S = \{x \in R^n : a^T x \leq k\}$
  - Polyhedron: $S = \{x \in R^n : Ax \leq b\} \subset R^n$
  - Intersection of convex sets: $S_1 \cap S_2$ is convex if S_1 and S_2 are convex sets.
  - Scaling and translation of a convex set: $aS + b$ is convex if S is convex.

Source: Bertsekas (2005)

Convex Functions

- Let $f: S \subset R^n \to R$ where S is a nonempty convex set. The function f is convex on S if for each $x, y \in S$ and for each $\alpha \in (0,1)$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y).$$



Source: Bertsekas (2005) and Bazaraa et al. (1993)

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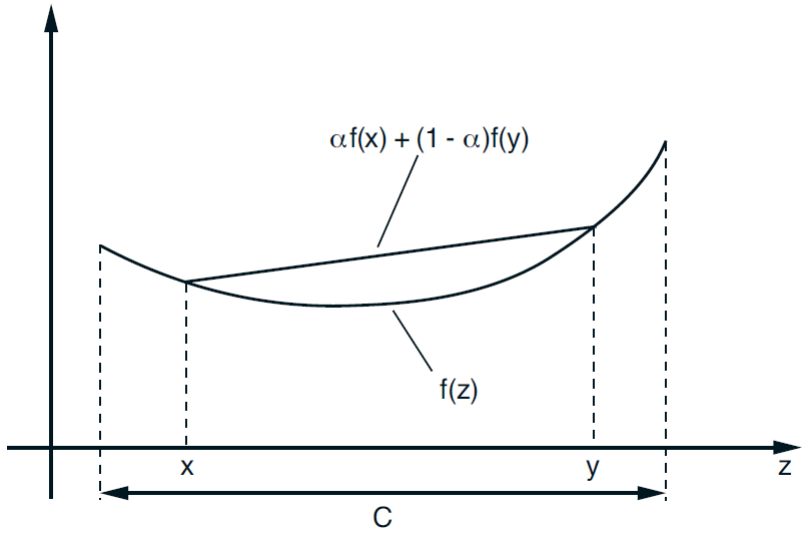
- Examples
  - $f(x) = 3x + 4$
  - $f(x) = |3x + 4|$
  - $f(x) = 3x^2 - 2x$
  - $f(x_1, x_2) = 2x_1^2 + 2x_2^2 - 2x_1 x_2$



- We say the function f on S is concave if $-f$ is convex on S.

Source: Bertsekas (2005) and Bazaraa et al. (1993)

Convex Functions

- Operations that preserve convexity

- Let $f_1, \dots, f_k : S \subset R^n \to R$ be convex functions on a convex set S. Then
  - Positive combination: $f(x) = \sum_{j=1}^{k} \alpha_j f_j(x)$, where $\alpha_j > 0$ for $j = 1, \dots, k$, is a convex function.
  - Maximization: $f(x) = \max\{f_1(x), \dots, f_k(x)\}$ is a convex function.

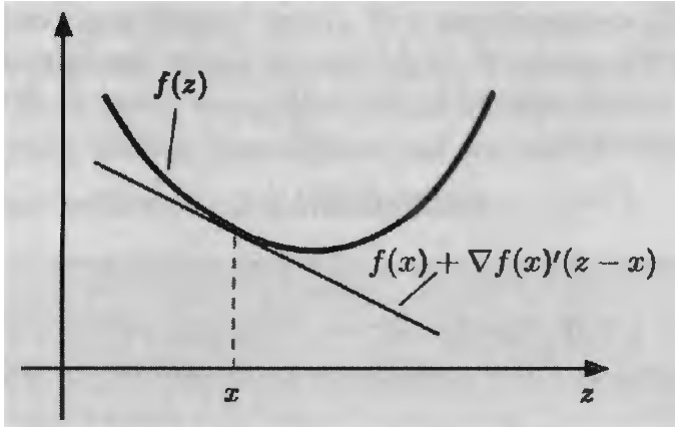
Source: Bertsekas (2005) and Bazaraa et al. (1993)

Differentiable Convex Functions

Proposition: Let $S \subset R^n$ be a convex set and let $f: R^n \to R$ be differentiable over R^n.
f is convex over S if and only if

$$f(z) \geq f(x) + \nabla f(x)^T (z - x), \forall x, z \in S.$$

- Proof



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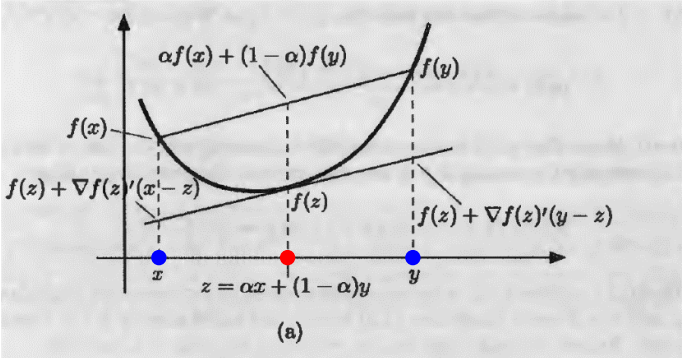
Choose any $x, y \in S$ and $\alpha \in [0,1]$, and let

$$z = \alpha x + (1 - \alpha) y.$$



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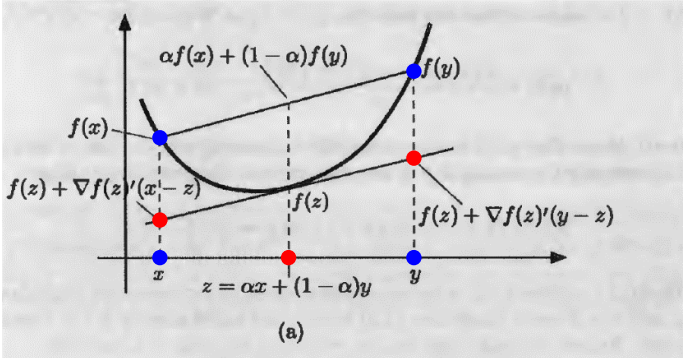
Proof

Choose any $x, y \in S$ and $\alpha \in [0,1]$, and let

$$z = \alpha x + (1 - \alpha) y.$$

Using the inequality in the proposition, we obtain

$$f(x) \geq f(z) + \nabla f(z)^T (x - z)$$
$$f(y) \geq f(z) + \nabla f(z)^T (y - z)$$



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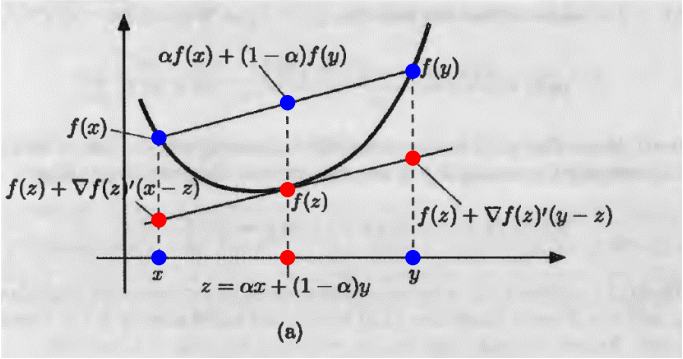
$$f(x) \geq f(z) + \nabla f(z)^T (x - z)$$
$$f(y) \geq f(z) + \nabla f(z)^T (y - z)$$

We have

$$\alpha f(x) + (1 - \alpha) f(y) \geq f(z) + \nabla f(z)^T (\alpha x + (1 - \alpha) y - z)$$
$$= f(z)$$

➔ f is convex.



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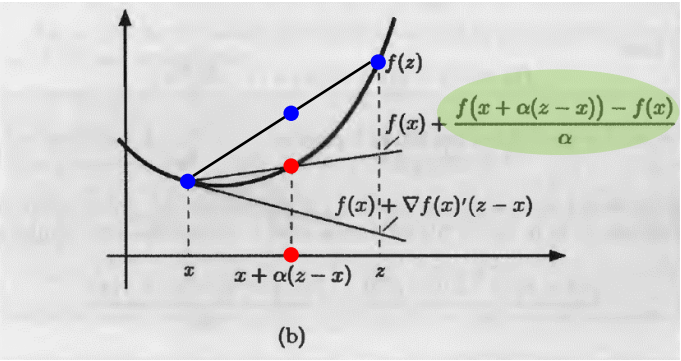
Proof

We have (for $0 < \alpha < 1$, due to convexity)

$$f(x + \alpha(z - x)) \leq \alpha f(z) + (1 - \alpha) f(x),$$

or

$$\frac{f(x+\alpha(z-x)) - f(x)}{\alpha} \leq f(z) - f(x). \qquad (**)$$



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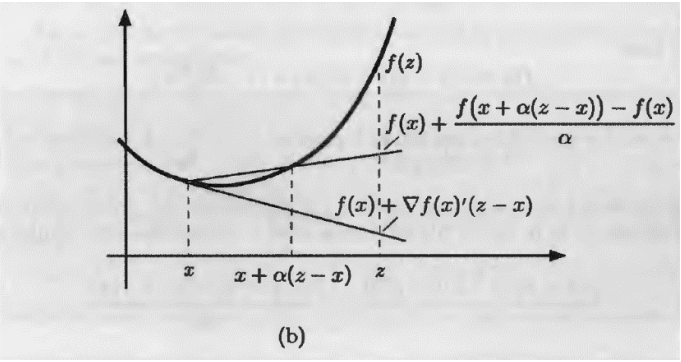
Proof

Consider any α_1, α_2 with $0 < \alpha_1 < \alpha_2 < 1$, and let

$$\bar{\alpha} = \frac{\alpha_1}{\alpha_2} < 1, \bar{z} = x + \alpha_2 (z - x). \qquad (*)$$



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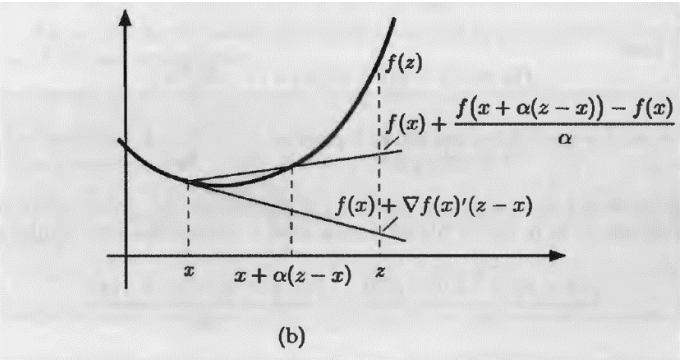

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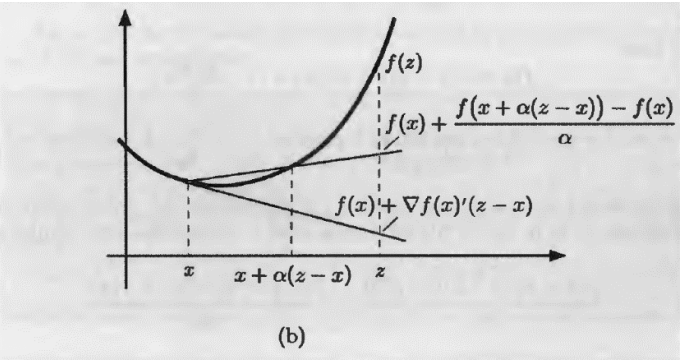
or

$$\frac{f(x+\bar{\alpha}(\bar{z}-x)) - f(x)}{\bar{\alpha}} \leq f(\bar{z}) - f(x). \qquad (**)$$

Substituting (*) in (**), we have

$$\frac{f(x+\alpha_1 (z-x)) - f(x)}{\alpha_1} \leq \frac{f(x+\alpha_2 (z-x)) - f(x)}{\alpha_2}.$$



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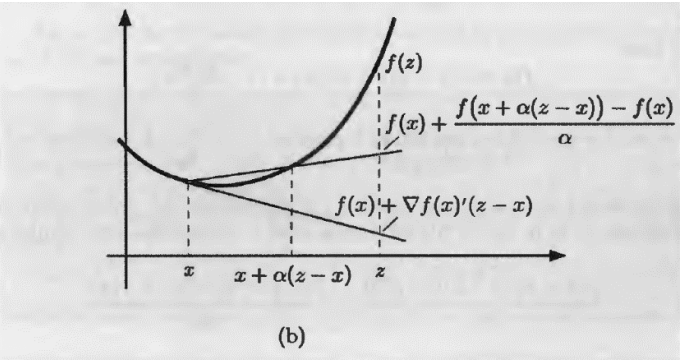
Proof

$$\frac{f(x+\alpha_1 (z-x))-f(x)}{\alpha_1} \leq \frac{f(x+\alpha_2 (z-x))-f(x)}{\alpha_2}.$$

Therefore, $f(x) + \frac{f(x+\alpha (z-x))-f(x)}{\alpha}$ lies below $f(z)$, is monotonically nonincreasing as $\alpha \to 0$, and converges to $f(x) + \nabla f(x)^T (z - x)$.

➔ $f(z) \geq f(x) + \nabla f(x)^T (z - x)$.



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Local and Global Optima

Local and Global Optima

- A vector x^* is an unconstrained local minimum of a function $f: R^n \to R$ if it is no worse than its neighbors; that is, if there exists an $\epsilon > 0$ such that

$$f(x^*) \leq f(x), \forall x \in R^n \text{ with } \|x - x^*\| < \epsilon \text{ or } x \in N_\epsilon(x^*).$$

- A vector x^* is an unconstrained global minimum of a function $f: R^n \to R$ if it is no worse than all other vectors; that is, if

$$f(x^*) \leq f(x), \forall x \in R^n.$$

- A vector x^* is a local minimum of $f: R^n \to R$ over X if $x^* \in X$ and there exists an $\epsilon > 0$ such that

$$f(x^*) \leq f(x), \forall x \in X \text{ with } \|x - x^*\| < \epsilon \text{ or } x \in N_\epsilon(x^*).$$

- A vector x^* is a global minimum of $f: R^n \to R$ over X if $x^* \in X$ and

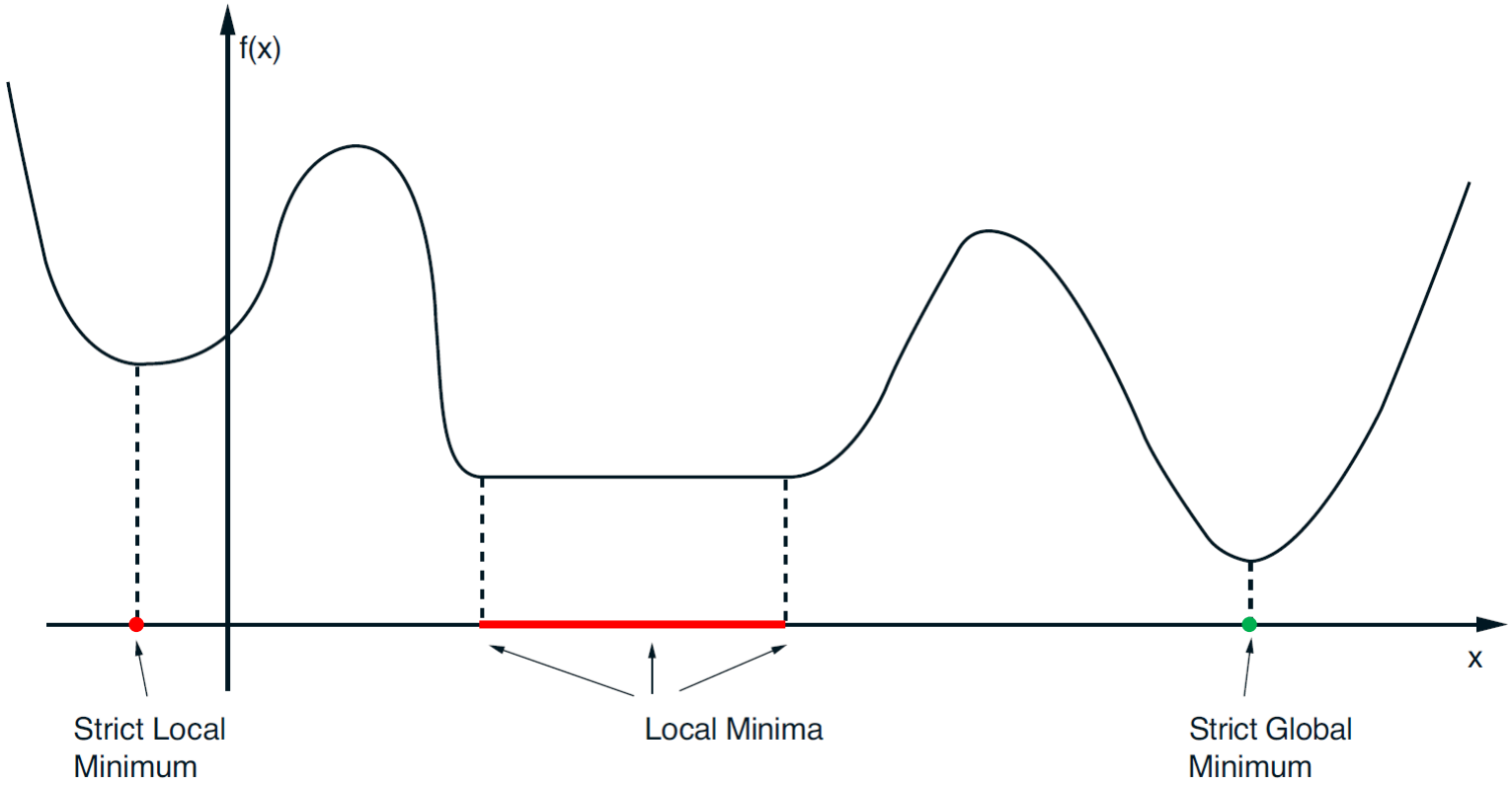
$$f(x^*) \leq f(x), \forall x \in X.$$

- The local/global minimum x^* is strict if the inequality is strict for $x \neq x^*$.

Source: Bertsekas (2005)

Local and Global Optima

- Unconstrained optimization



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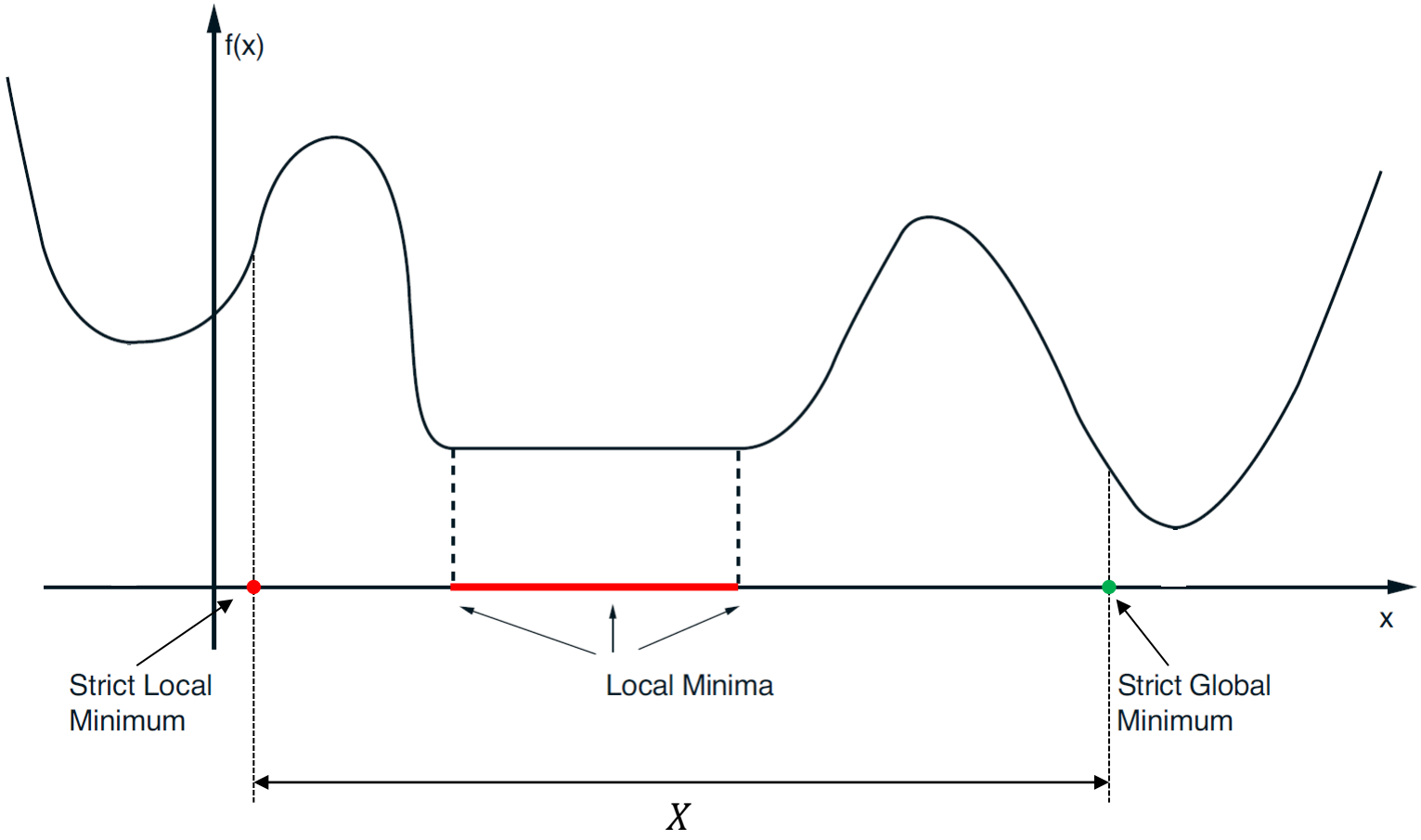
*Local and Global Optima

- Constrained optimization over X



Source: Bertsekas (2005)

Local and Global Optima

Theorem: If $f: R^n \to R$ is convex over a convex set $X \subset R^n$, then a local minimum of f over X is also a global minimum.

- Proof

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Theorem: If $f: R^n \to R$ is convex over a convex set $X \subset R^n$, then a local minimum of f over X is also a global minimum.

Proof (by contradiction)

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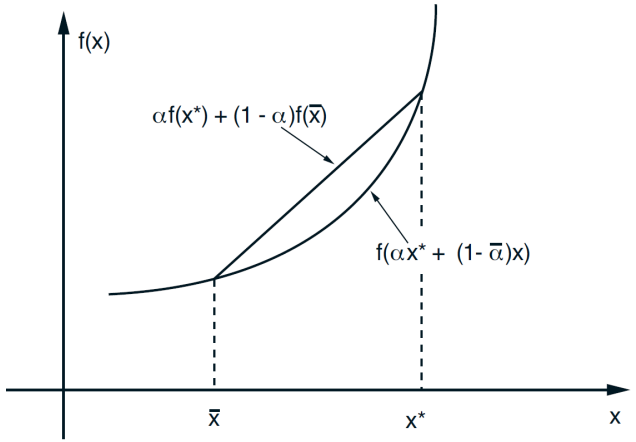
Then there exists some $\bar{x}$ such that $f(\bar{x}) < f(x^*)$.



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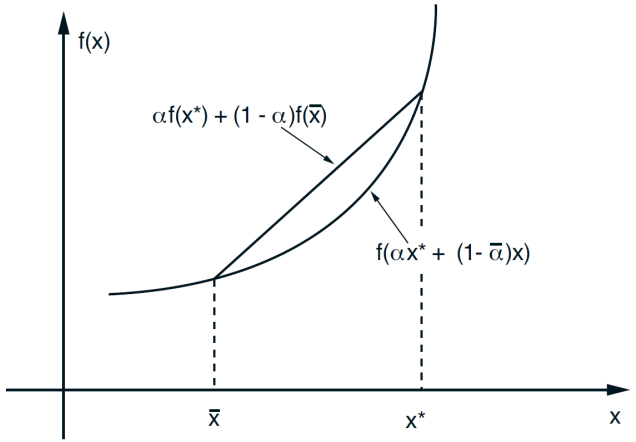

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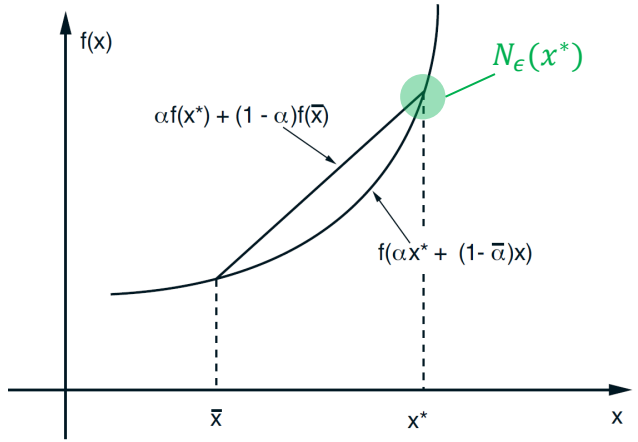
Then there exists some $\bar{x}$ such that $f(\bar{x}) < f(x^*)$.

Using the convexity of f, we have

$$f(\alpha x^* + (1-\alpha)\bar{x}) \leq \alpha f(x^*) + (1-\alpha) f(\bar{x}) < f(x^*), \forall \alpha \in (0,1).$$

With a sufficiently large $\alpha > 0$, $\alpha x^* + (1-\alpha)\bar{x} \in N_\epsilon(x^*)$.

This contradicts the local minimum of x^*.



Source: Bertsekas (2005)

Unconstrained Optimization

Unconstrained Optimization

$$\boxed{\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in R^n. \end{array}}$$

- If function f is differentiable, consider small variations Δx from a given x^*

*Unconstrained Optimization

$$\boxed{\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in R^n. \end{array}}$$

- If function f is differentiable, consider small variations Δx from a given x^*
  - First order approximation of the cost variation
$$f(x^* + \Delta x) - f(x^*) \approx \nabla f(x^*)^T \Delta x$$
  - Second order approximation of the cost variation
$$f(x^* + \Delta x) - f(x^*) \approx \nabla f(x^*)^T \Delta x + \frac{1}{2} \Delta x^T \nabla^2 f(x^*) \Delta x$$
  - We expect that if x^* is an unconstrained local minimum, the first order cost variation due to a small variation Δx is nonnegative, i.e.,
$$\nabla f(x^*)^T \Delta x \geq 0 \rightarrow \nabla f(x^*)^T = 0$$
  - We also expect that the second order cost variation due to Δx is nonnegative, i.e.,
$$\nabla f(x^*)^T \Delta x + \frac{1}{2} \Delta x^T \nabla^2 f(x^*) \Delta \mathrm{x} \geq 0 \rightarrow \Delta x^T \nabla^2 f(x^*) \Delta x \geq 0, \forall \Delta x \in R^n$$
$$\rightarrow \nabla^2 f(x^*) \text{ positive semidefinite}$$

Optimality Conditions

Proposition: Let x^* be an unconstrained local minimum of $f: R^n \to R$, and assume that f is continuously differentiable in an open set S containing x^*. Then

$$\nabla f(x^*) = 0. \qquad \text{(First order necessary condition)}$$

If in addition f is twice continuously differentiable within S, then

$$\nabla^2 f(x^*)\text{: positive semidefinite.} \qquad \text{(Second order necessary condition)}$$

Source: Bertsekas (2005)

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If in addition f is twice continuously differentiable within S, then

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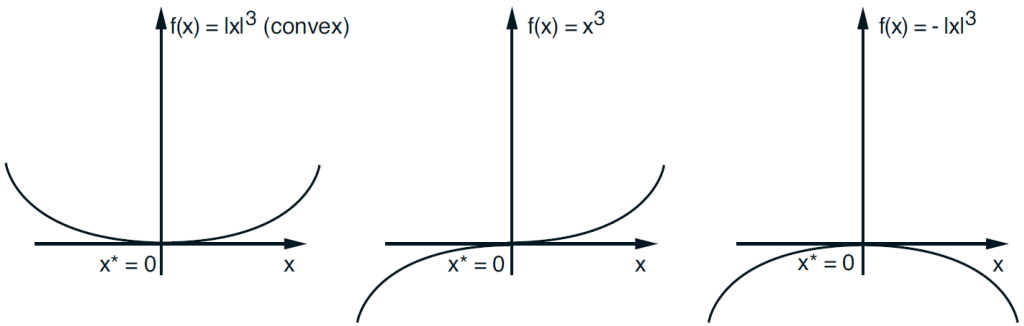
- First order condition: zero slope at a local minimum x^*
- Second order condition: nonnegative curvature at a local minimum x^*
- Note: There may exist points that satisfy both first and second order conditions but are not local minima.



Source: Bertsekas (2005)

Optimality Conditions: The Convex Case

Proposition: Let $f: R^n \to R$ be a convex function over a convex set $X \subset R^n$.

a) If f is continuously differentiable, then

$$\nabla f(x^*)^T (x - x^*) \geq 0, \forall x \in X,$$

is a necessary and sufficient condition for $x^* \in X$ to be a global minimum of f over X.

b) If X is open and f is continuously differentiable over X, then $\nabla f(x^*) = 0$ is a necessary and sufficient condition for $x^* \in X$ to be a global minimum of f over X.

Source: Bertsekas (2005)

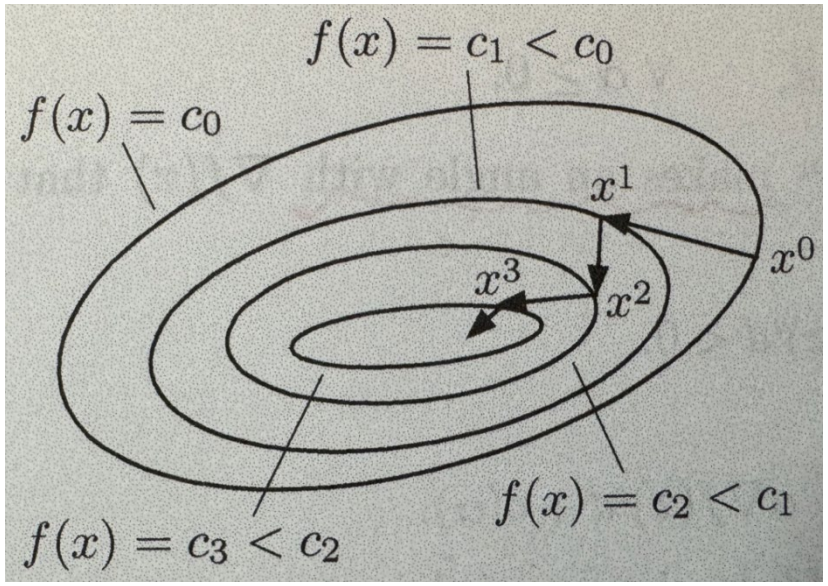
Gradient Methods

Iterative Descent Methods

- Algorithm outline
  - Start at some initial point x^0
  - Successively generate x^1, x^2, …, such that f is decreased at each iteration, i.e.,

$$f\left(x^{k+1}\right) < f(x^k),\ k = 0, 1, \ldots$$

  - Hope to decrease f all the way to its (local) minimum.



Source: Bertsekas (2016)

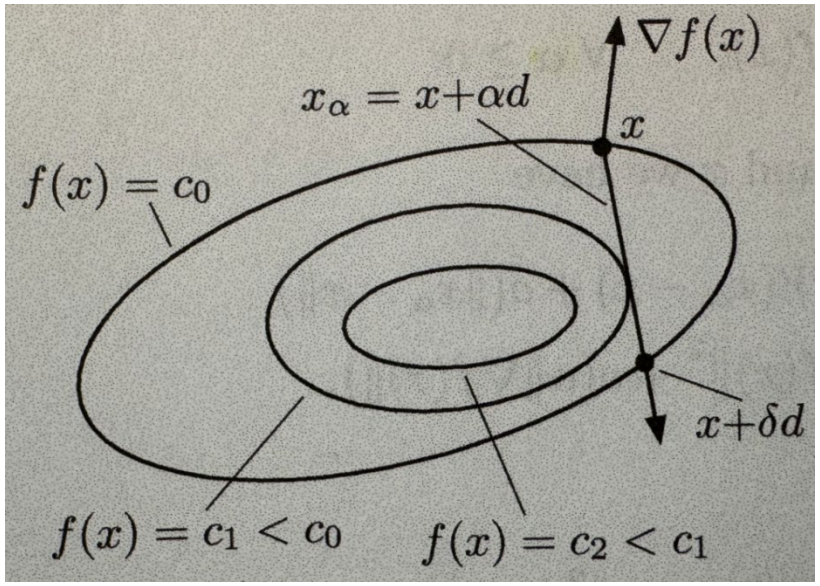
Gradient Methods

- Given $x \in R^n$ with $\nabla f(x) \neq 0$
- Select a direction $d \in R^n$ makes an angle with $\nabla f(x)$ that is greater than 90 degree, i.e., $\nabla f(x)^T d < 0$.
- We have (by the first-order Taylor's expansion)

$$f(x + \alpha d) = f(x) + \alpha \nabla f(x)^T d + o(\alpha),$$

- For α near zero, the term $\alpha \nabla f(x)^T d$ dominates $o(\alpha)$.
- Therefore, for positive but sufficiently small α,

$$f(x + \alpha d) < f(x).$$



Gradient Methods

Algorithm 6 Gradient Method

Step 0. (Initialization) Start at some initial point x^0.

Step 1. At each iteration k ($k = 0, 1, \ldots$),

If $\nabla f(x^k) = 0$ (or $\|\nabla f(x^k)\| \leq \epsilon$), terminate with x^k being the local minimum.

If $\nabla f(x^k) \neq 0$,

Select the direction d^k so that $\nabla f(x^k)^T d^k < 0$ and stepsize $\alpha^k > 0$.

Let $x^{k+1} = x^k + \alpha^k d^k$.

Go to Step 1.

Gradient Methods

- Selection the descent direction
  - Many gradient methods are specified in the form
$$x^{k+1} = x^k - \alpha^k D^k \nabla f(x^k),$$
  - where D^k is a positive definite symmetric matrix.
  - Note that $d^k = -D^k \nabla f(x^k)$, so that
$$\nabla f(x^k)^T d^k = -\nabla f(x^k)^T D^k \nabla f(x^k) < 0.$$

Steepest Descent Method

- Selection the descent direction
  - Many gradient methods are specified in the form

$$x^{k+1} = x^k - \alpha^k D^k \nabla f(x^k),$$

  - where D^k is a positive definite symmetric matrix.
  - Note that $d^k = -D^k \nabla f(x^k)$, so that

$$\nabla f(x^k)^T d^k = -\nabla f(x^k)^T D^k \nabla f(x^k) < 0.$$

  - Steepest descent: $D^k = I$ ➔ $d^k = -\nabla f(x^k)/\|\nabla f(x^k)\|$
    - Simplest but often slow convergence

  - Newton's method: $D^k = \left(\nabla^2 f(x^k)\right)^{-1}$ (provided that $\nabla^2 f(x^k)$ is positive definite)
    - Often fast convergence but computational expensive

Source: Bertsekas (2005)

Gradient Methods

- Stepsize selection
  - Constant stepsize: $\alpha^k = s > 0$
    - s too large, may diverge
    - s too small, slow convergence
  - Diminishing stepsize: $\alpha^k \to 0$ and $\sum_k^\infty \alpha^k = \infty$
    - Descent property may only holds when stepsize becomes sufficiently small.
    - The condition $\sum_k^\infty \alpha^k = \infty$ guarantees that $\{x^k\}$ does not converge to a nonstationary point.
    - Good theoretical convergence properties
  - Minimization rule: $\alpha^k \in \arg\min_\alpha f(x^k + \alpha d^k)$
  - Limited minimization rule: $\alpha^k \in \arg\min_{\alpha \in [0,s]} f(x^k + \alpha d^k)$
    - Typically implemented with the aid of one-dimensional line search algorithms, e.g., bisection method, golden section method, etc.

Q & A